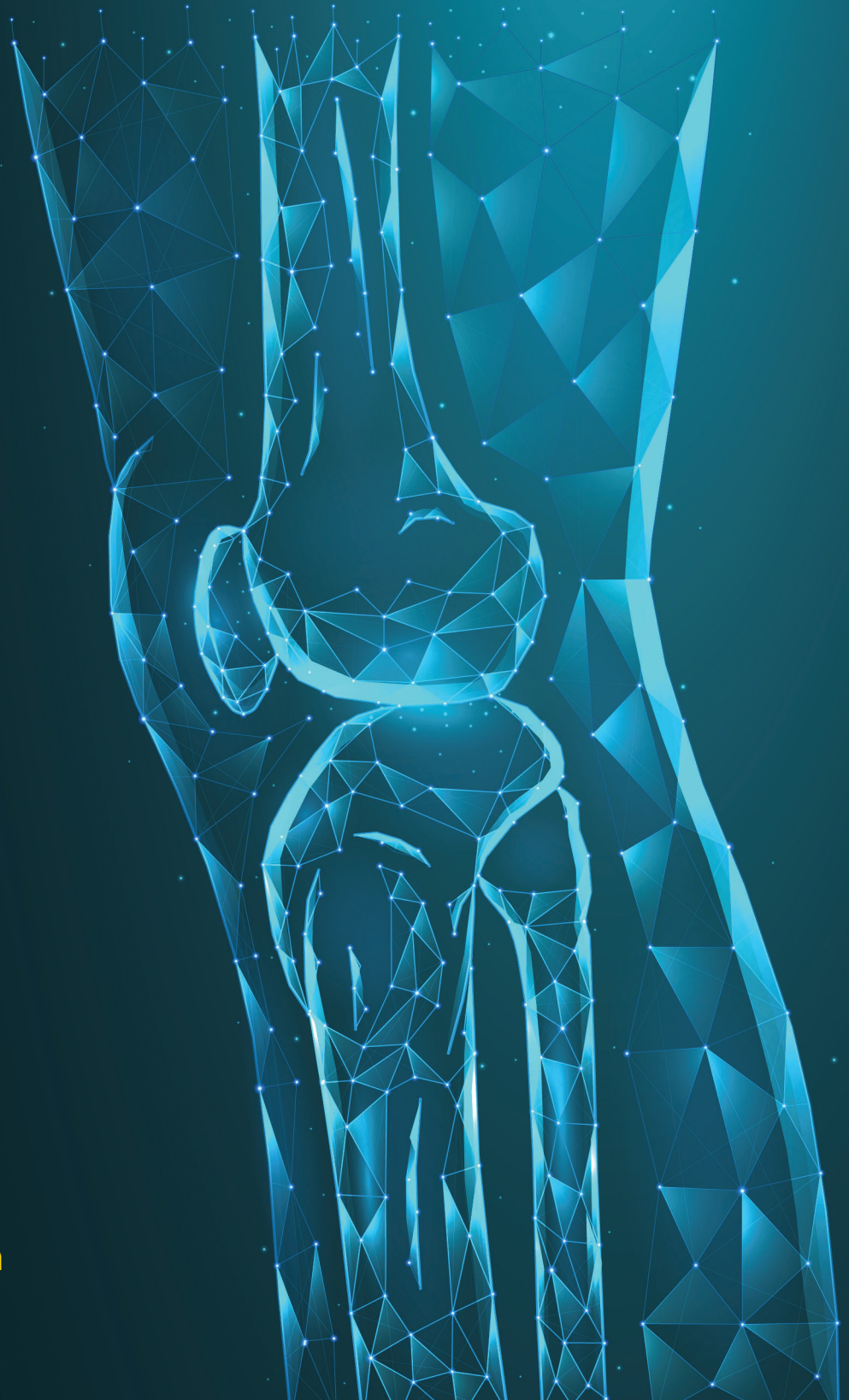


Aspetar ACL

Rehabilitation Protocol



2023-24 Edition

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Introduction

In Aspetar it is our mission to assist all athletes achieve their maximum performance and full potential. Our vision is to be a global reference centre in sports medicine. Specific to Anterior Cruciate Ligament (ACL) rehabilitation we aim to help every athlete, regardless of their participation level, get back to their desired sport and activity as quickly and safely as possible. In addition, we want to advance the standard of ACL rehabilitation world wide. Through continuous review and enhancement of our clinical practice, pathways, and research, we aim to improve rehabilitation strategies and outcomes after ACL injury.



Aspetar, Orthopaedic and Sports Medicine Hospital

We individualise our rehabilitation program to address the needs and deficits of each athlete, ensuring a successful outcome as efficiently as possible.

We define success after an ACL injury as:

- Returning to the desired level of sporting participation or function
- Minimizing the risk of subsequent injury to either knee
- Ensuring the absence of any ongoing pain, swelling, or morbidity in the injured knee
- Optimising physical function to pre-injury levels while addressing any injury specific risk factors or performance deficits
- Achieving a return to pre-injury levels of performance or better

The Aspetar ACL Rehabilitation Protocol outlines the key physical qualities that need to be addressed during rehabilitation. It specifies when these qualities should be introduced, highlights key milestones along the journey, and establishes criteria marking the completion of rehabilitation and the return to baseline sporting participation. There is no single best way or approach to rehabilitate an athlete after an ACL injury. Our previously published Clinical Practice Guidelines in the BJSM¹ detail the research supporting various modalities used in the process. There is a huge variety of exercise selection, coaching and programming methods, that can all achieve similar results and outcomes. The key to the process, and the aim of this protocol, is to ensure consistency among all parties involved in rehabilitation regarding the key performance indicators assessed and addressed throughout the process. This consistency guarantees a unified approach when introducing and achieving crucial milestones in the rehabilitation journey.

This protocol undergoes constant review and evolution to align with advancements in research and clinical practice. It serves as a roadmap for both clinicians and athletes, facilitating their rehabilitation journey and ensuring a safe and efficient arrival at the endpoint.

Our **protocol** specifies the timing for initiating each component of our rehabilitation program, outlines the starting point for each component, and illustrates how these components integrate.

Our **curriculum** offers exercises and progressions designed to transition our athletes from surgery to discharge as quickly, safely, and consistently as possible.

Our **guidelines** dictate which tools should be incorporated into the rehabilitation protocol according to research findings, to effectively achieve our goals.



Summary of the Recommendations on Rehabilitation After ACLR¹

Core Principles

Every ACL Rehabilitation protocol should have several core principles to ensure consistency of approach and outcome.

Our core principles, which are embodied in this protocol, include:

- **Individualised approach** – Each phase of rehabilitation and progression through it is individualised to address the athlete's specific deficits and their response, adaptation, and improvement during that training block. There is variability in graft selection and comorbidity to the knee, previous injury history, previous training history, baseline physical competency and response to training stimulus. Therefore, it is essential to tailor each phase to the athlete's current physical status to optimise progression and outcomes.
- **Assessment guided rehabilitation** – All athletes should undergo periodised clinical and physical assessment to track progress, prioritize ongoing physical deficits, and optimise goal setting and programming. This will facilitate reflection on the effectiveness of the previous rehabilitation phase and ensure that the athlete is ready for transition to the next phase safely.



Riadh Assessment and Movement Analysis Laboratory (RAMAL)

- **Address multiple physical qualities concurrently** – While various physical qualities will be prioritised at different phases of rehabilitation, it is essential that each phase complements and transitions effectively into the next. Consequently, the rehabilitation program must focus on addressing priority deficits while also introducing exposure to additional qualities essential for later phases of rehabilitation.
- **Focus on quality as well as quantity of training stimulus** – While training frequency and intensity are important variables, prioritising “practice” over “training” during rehabilitation emphasises movement quality as much as movement quantity. This approach optimises motor learning and development, ensuring appropriate biomechanical and physiological adaptations are achieved, while minimizing potential knee aggravation, for the time and effort invested.
- **Multidisciplinary approach centred around the athlete and their goals** – Many disciplines play significant roles in the rehabilitation process, including orthopaedic surgeons, physiotherapists, strength and conditioning coaches, nutritionists, psychologists, and sports specific coaches among others.² The key to a successful multidisciplinary team is alignment in roles and responsibilities, with clarity provided by the rehabilitation protocol regarding criteria for each phase of rehabilitation and the transition from rehabilitation to unrestricted sporting participation.



Multidisciplinary Team (with permission from Aspetar Sports Medicine Journal)²

Aspetar ACL Rehabilitation Team

Name	Role
Riadh Miladi	Director of Rehabilitation Department
Bruna Antunes	Physiotherapist
Olivia Barbosa	Physiotherapist
Bader Belfekih	Physical Coach
Andreas Bjerregaard	Physiotherapist
Brahim Boubaker	Physiotherapist
Brendan Butler	Physiotherapist
Mayolo Camacho	Physiotherapist
Tim Campkin	Physiotherapist
Admir Cehic	Physiotherapy Assistant
Anis Chaouachi	Physical Coach
In Hyuk	Physiotherapist
Konstantinos Defteraios	Physiotherapist
Luke Heath	Senior Physiotherapist
Anissa Fandri	Senior Physical Coach
Oana Gheorghita	Physiotherapist
Abdallah Itani	Lead Clinical Physiotherapist
Shaun Jankielsohn	Physiotherapist
Ahmed Al Jawad	Physiotherapist
Enda King	Head of Elite Performance and Development
Gergely Kis	Physical Coach
Roula Kotsifaki	Head of Assessment Lab
Nicoletta Luchini	Senior Physiotherapist
Calum MacAskill	Physiotherapist
Hassine Manai	Senior Physiotherapist
Dustin Maree	Physiotherapist
Joao Marques	Senior Research Analyst
Cristian Nae	Physiotherapist
Nuno Nascimento	Physiotherapist
Mansour Otayek	Physiotherapist
Theodosia Palli	Sports Rehabilitator
Hristina Petrova	Physiotherapist
Gareth Robinson	Physiotherapist
Vasileios Sideris	Senior Biomechanist
Toni Jane Snoxell	Physiotherapist
Hassen Soltani	Physiotherapist
Sarah Talib	Physiotherapist
Christopher Thomas	Physical Coach
Cristina Vizante	Physiotherapist
Rod Whiteley	Assistant Director of Rehabilitation
Elaine Zammit	Physiotherapist

Multidisciplinary Team Approach

Every multidisciplinary team varies in size, composition, and available expertise. However, regardless of the team's makeup, it is essential to clearly outline the roles and responsibilities, initiation points, and transitions between team members.

Multidisciplinary Team Responsibilities

Within Aspetar, our ACL multidisciplinary team is structured to share responsibilities throughout the rehabilitation process as follows:

Aspetar ACL Multidisciplinary Team

Orthopaedics	Surgery and post-op precautions	Review of clinical progress	Review of clinical progress	Review of clinical progress	Review of clinical discharge criteria
Physiotherapy	Clinical/ROM Motor control Strength	Clinical/ROM Motor control Strength	Clinical/ROM Motor control Reactive strength Running mechanics	Clinical/ROM Motor control Reactive strength Running mechanics COD mechanics	Clinical/ROM Motor control Reactive strength Running mechanics COD mechanics
Hydrotherapy	Clinical/ROM Motor control Strength	Clinical/ROM Motor control Strength	Recovery	Recovery	Recovery
Conditioning	Upper body	Non impact	Low impact	On feet	On field
Strength			Strength Explosiveness	Strength Explosiveness	Strength Explosiveness
Sports Specific			Isolated skills	Closed drills	Open drills
Nutrition	Recovery Body composition Nutrient/energy deficiency	Recovery Support Hypertrophy	Support Hypertrophy Reduce muscle fatigue	Support strength and cardiovascular development	Support sports specific demands for performance
Psychology	Education Coping skills	Goals setting	Sleep and stress management	Motivation and compliance	Apprehension/fear Confidence in performance
Assessment	Test 1 > 6 Weeks	Test 2 > 12 Weeks	Test 3 > 18 Weeks	Test 4 > 24 Weeks	Test 5 > 30 Weeks

Multidisciplinary Team Transitions

The timeline and criteria for the introduction/transition to each component of the multidisciplinary team is as follows:

Orthopaedic Surgeon – from initial diagnosis

Physiotherapy – immediate post injury, pre-operative and immediate post-surgery

Nutrition – pre-operative

Psychology – pre-operative

Hydrotherapy – from week 3-4, once wound is healed

Conditioning – from week 3-4

Assessment Lab – pre-operative and from 6 weeks post-surgery

Strength – from week 12

- Full range open chain quadriceps and hamstring strengthening pain free
- Full knee extension and minimum 130° knee flexion
- Single leg squat to 90° knee flexion
- Front squat to 90° knee flexion
- Landing exercises commenced

Sports Specific Training – from week 18

- Pain and swelling free
- Full knee extension and minimum 130° knee flexion
- Single leg squat 90° knee flexion with neutral alignment
- Forward/backward cone hopping
- Neutral landing mechanics in submaximal hop in all three planes
(forward/backward, medial/lateral, 90° rotation clockwise/anti-clockwise)
- Running 16km/h x 200m x 8 repetitions without reaction (may be modified for court sport athletes)
- Symmetrical lateral push off and crossover step
- Isokinetic quadriceps and hamstring strength > 80% Limb Symmetry Index (LSI)
- Double leg countermovement and drop jumps concentric and eccentric impulse >80% LSI
- Single leg drop jump Reactive Strength Index (RSI) >80% LSI

Key Performance Indicators (KPIs)

The ongoing monitoring of the athlete's progress is essential to:

- Assess the response of their knee to the training stimulus
- Assess the change in key metrics across a training block
- Identify ongoing deficits that should be addressed in the next training block

Assessment and monitoring occurs formally or informally in nearly every interaction with the athlete however it is important to record KPIs on a structured basis to inform decision making.

Daily KPIs

Pain – NRS, location and activity

Knee ROM – flexion, extension, tibial internal rotation

Swelling – stroke test

Quads Activation/Lag – (less relevant in later stages of rehabilitation, unless there has been a flare up)

Weekly KPIs

Pain – NRS, location and activity

Knee ROM – flexion, extension, tibial internal rotation

Swelling – stroke test

Quads Activation/Lag – (less relevant in later stages of rehabilitation unless there has been a flare up)

Thigh Mass – circumference at 5cm and 10cm suprapatellar

Motor Control Progress – (SL squat, SL calf raise, front squat, SL romanian deadlift)

6 Weekly KPIs (by the Assessment Lab, appropriate for the stage of rehabilitation)

Pain – NRS, location and activity

Knee ROM – flexion, extension

Swelling – stroke test

Patient Reporting Outcomes – IKDC, Tampa Scale of Kinesiophobia

Motor Control – double leg squat, single leg squat, step down

Strength – isokinetic knee, handheld dynamometer ABD/ADD/ER, isometric ankle plantar-flexors

Explosiveness – double and single leg countermovement jump, single leg hop for distance

Reactive Strength – double and single leg drop jump

Running Biomechanics – treadmill

Change of Direction Biomechanics – 90° planned and unplanned

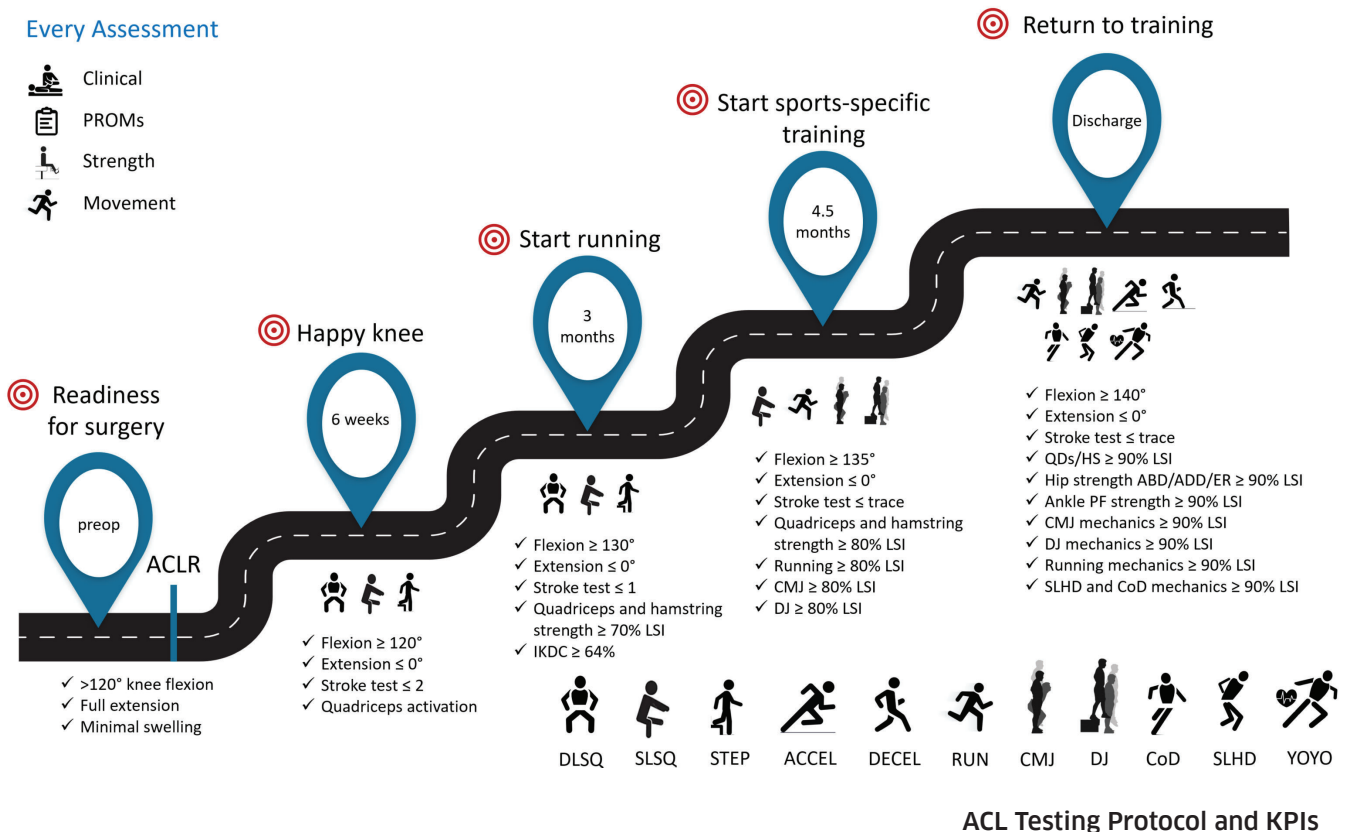
Conditioning – hand crank, submaximal ramp bike test, Yo-Yo

Assessment Lab Testing Protocol

Monitoring progress during rehabilitation is crucial to address any remaining deficiencies in an athlete's physical and psychological condition. Deficiencies can manifest in various areas such as strength, range of motion, joint laxity, performance, functional ability, and psychological readiness. To accurately identify these areas of improvement, it is essential to use tests and metrics with sufficient sensitivity. Continuous monitoring of these metrics during the rehabilitation process enables athletes to track their improvement and ensure a comprehensive recovery that addresses all aspects affected by the ACL injury and subsequent intervention.

Despite the importance of a successful return to performance after surgery and rehabilitation, not all athletes regain their pre-injury performance levels upon return to their sport. This highlights the need for objective performance metrics at the time of return to sport to enhance secondary prevention and accurately evaluate an athlete's readiness to perform at their full potential.

Based on extensive clinical experience and observed deficits in athletes during their return to sport, a comprehensive testing battery has been developed to assess various aspects crucial for a successful return to sport after ACL reconstruction.



Assessment Lab Tests and Selected Metrics

Outcomes	Tests	Metrics
Clinical	Active knee flexion	Range of motion
	Passive knee extension	Extension deficit
	Swelling - stroke test	0-3
	Stability: Lachman Pivot shift Instrumented laxity test Subjective laxity	End feel 0-3 Difference between limbs in translation and rotation 0-10
Patient Reported Outcome Measures	Pain at rest	0-10 (Numeric rating scale)
	Pain during activity	0-10 (Numeric rating scale)
	IKDC	0-100%
	Tampa scale of kinesiophobia	11-44
Flexibility-ROM	Weight-bearing lunge test (Ankle dorsiflexion)	Distance toe-to-wall
Strength	Quadriceps and hamstring	Quadriceps peak torque isometric 60° Hamstring peak torque isometric 30° Quadriceps peak torque concentric 60°/s Hamstring peak torque concentric 60°/s Hamstring peak torque eccentric 60°/s
	Hip	Abductors peak force eccentric (break test) Adductors peak force eccentric (break test) External rotators peak force isometric
	Ankle plantar-flexors	Peak isometric force

Assessment Lab Tests and Selected Metrics

Outcomes	Tests	Metrics
Movement	Double leg squat	- Concentric peak force LSI - Eccentric impulse LSI
	Single leg squat	Knee ROM
	Step down	Peak landing force LSI
	Countermovement jump	- Jump height - Eccentric impulse LSI - Concentric impulse LSI - Peak landing force LSI
	Single leg countermovement jump	- Jump height LSI - Eccentric impulse LSI - Concentric impulse LSI
	Double leg drop jump	- Jump height - Contact time - RSI - Eccentric impulse LSI - Concentric impulse LSI - Peak landing force LSI
	Single leg drop jump	- Jump height LSI - Contact time LSI - RSI LSI - Eccentric impulse LSI - Concentric impulse LSI
	Running	- Ankle/knee/hip joint angles and moments in three planes - Ground reaction forces and ankle/knee/hip joint power
	Single leg hop for distance	- Ankle/knee/hip joint angles and moments in three planes - Ground reaction forces and ankle/knee/hip joint power
	Change of direction 90°	- Ankle/knee/hip joint angles and moments in three planes - Ground reaction forces and ankle/knee/hip joint power
Cardiovascular /Endurance	Yo-Yo test	- Total distance - Heart rate - Maximal aerobic speed

Pre-Operative Criteria

The restoration of homeostasis to the knee prior to surgery is essential for the short and long term success of any surgical intervention. It is well recognised that performing early surgery on a knee with ongoing range of motion deficits and knee effusion significantly increases the likelihood of persistent stiffness, post-surgery inhibition, and the development of arthrofibrosis. The preoperative period also offers an opportunity to familiarize the athlete with many of the exercises and interventions (such as cryotherapy, NMES, etc.) that they will use in the immediate post-operative period, enabling them to “hit the ground running” immediately after surgery.

The essential preoperative criteria before any surgical intervention are:

- Full/symmetrical knee extension
- >120° knee flexion
- Minimal swelling
- No quadriceps lag
- Normal gait

Exceptions to these criteria may occur in cases of concurrent meniscal or chondral injuries preventing the achievement of these goals, or in larger knee injuries involving multi-ligament damage that may have neurological or vascular compromise requiring urgent intervention.

Post-Operative Precautions and Indications

The primary precautions after ACL reconstruction relate to knee range of motion and weight bearing status. Postoperative precautions are determined by the orthopaedic surgeon based on the area of the pathology and the specific procedure performed for each individual athlete. The baseline case in this protocol assumes ACL reconstruction using Bone-Patellar Tendon-Bone, Hamstring, and Quadriceps grafts without concurrent injuries requiring repair/reconstruction, which would have no restrictions on ROM (as tolerated) or weight bearing (as tolerated) post-surgery.

The baseline case would usually include two additional functional precautions at the surgeon's discretion:

- Kneeling Prayer: Modified from month 4, full knee flexion from month 6
- Driving: Right knee: 4-6 weeks // Left knee: 2-3 weeks

Concomitant injuries or procedures often dictate weight bearing and early range of motion precautions. In the absence of clear direction from the surgeon upon initial referral, there should be an automatic default, which should be confirmed or modified with the surgeon on subsequent communication:

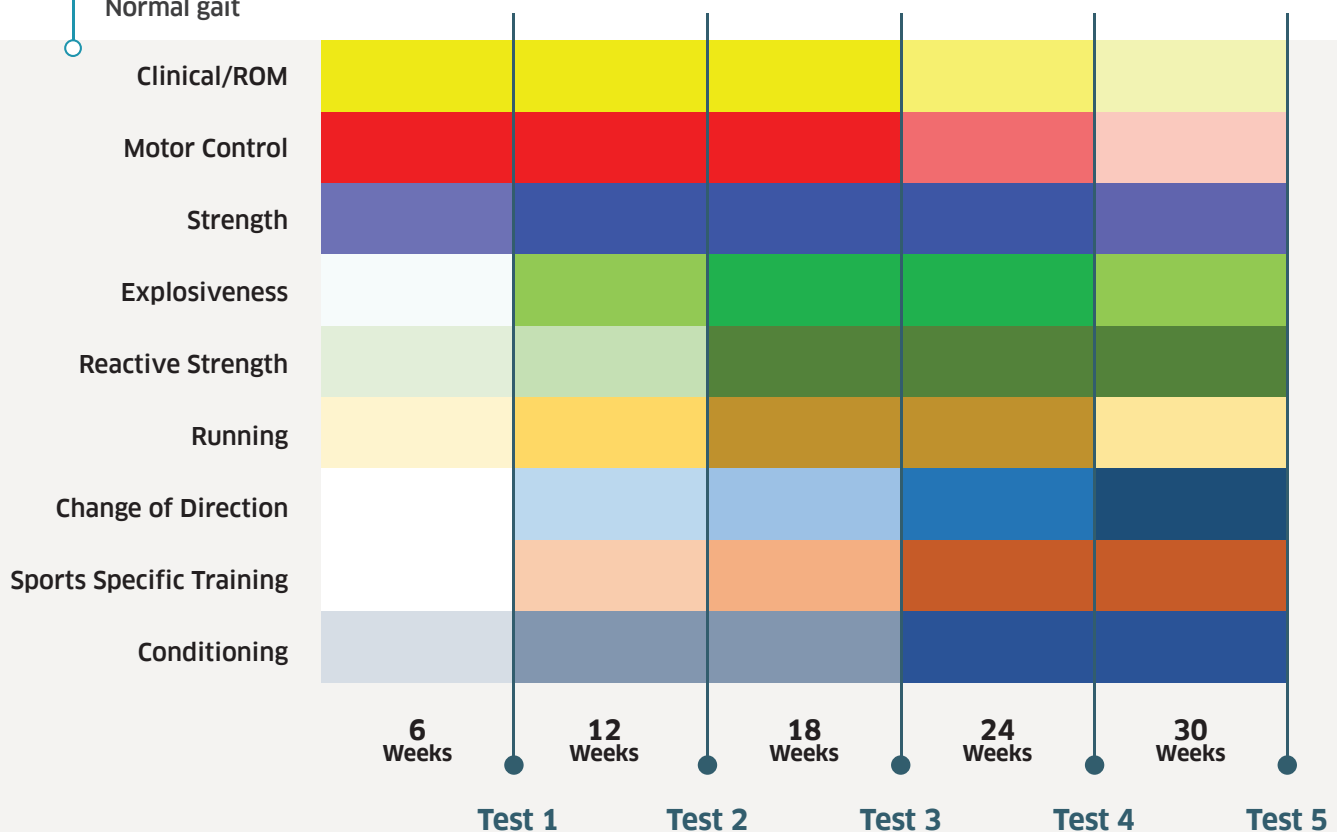
- Meniscal repair: Non-weight bearing (NWB) for 4 weeks and braced 0° to 90°
- Chondral repair: NWB for 4 weeks and braced 0° to 90°
- Lateral Extra-articular Tenodesis: Weight bearing and ROM as tolerated
- Secondary ligament injuries (i.e posterolateral corner): NWB for 4 weeks and braced 0° to 90°

Components of Rehabilitation

As outlined in the Core Principles section, one of the keys to successful rehabilitation after ACL reconstruction is developing concurrent physical qualities at appropriate stages of rehabilitation. This ensures that the athlete is adequately prepared for each transition throughout the rehabilitation process and that all key competencies are achieved in a timely manner before discharge.

Pre-op Criteria

Full extension
>120° knee flexion
Minimal swelling
No quadriceps lag
Normal gait



Patient Specific Discharge Criteria

Competitive athletes
Recreational athletes

The role of a protocol is not to dictate which exercises, programming, and periodization methods to use, as there are numerous approaches to achieving the desired outcomes. Instead, its purpose is to outline all the key criteria that should be included and to suggest the entry and minimum exit points for each component.

Clinical/ROM

In the early phase immediately post surgery, the goals are to **protect** the knee to facilitate recovery, restore appropriate **muscle activation/minimize muscle loss**, and **normalise gait** while regaining knee range of motion. All interventions should be guided by the post-surgical precautions outlined by the surgeon and should be introduced and progressed based on how the knee symptoms respond. The rate of progress may vary for each athlete, but every individual should see weekly improvements across these measures, especially in ROM. Any plateau in the first six weeks may be cause for review with their surgeon.

Protection

Post-op Precautions	As per surgeon's instructions
Wound Care	Avoid pool/hydrotherapy until wound fully healed
Swelling/effusion	Compression garments and regular cryotherapy daily

Muscle Activation

Isometric contraction quadriceps and hamstring	Important to isolate good quality inner-range quadriceps activation and closed chain hamstring activation in first two weeks
Neuromuscular Electrical Stimulation (NMES)	Can be used daily and ideally with active contraction
Surface Electromyography (EMG)	Can be used daily to improve independent muscle activation and improve motor learning
Blood Flow Restriction (BFR) Training	Can be included 3-4 times per week once no contraindications

ROM/Gait

Gait Re-education	Progress from 2 to 1 crutch to full weight bearing over a two week period as tolerated
Balance	Progress from double to tandem to single leg balance exercises as tolerated
Knee Flexion	Progress active knee flexion as tolerated while concurrently addressing knee swelling and any patellofemoral hypomobility
Knee Extension	Achieve and retain symmetrical knee extension in the first week post surgery



EMG and Active Hamstring Contraction



NMES Quadriceps



Knee Flexion ROM/PF Mobility

Motor Control/Isolated Muscle Strength (excluding quadriceps/hamstring)

Exercise Streams	Entry Point Criteria	Suggested Entry Exercise	Minimal Exit Point
Double leg squat	Once partial weight bearing	Goblet squat	Front squat
Double leg hinge/deadlift	Once partial weight bearing	Mid thigh hinge/deadlift	Single leg RDL criteria
Single leg squat	Once full weight bearing	Total gym/leg press	Single leg squat to box at knee height, with neutral frontal plane alignment
Single leg hinge/SLRDL	Once full weight bearing	Double leg hinge/deadlift	Single leg RDL with trunk Rotation
Single leg calf raise	Once full weight bearing	Double leg calf raise	Single leg calf raise onto 1st MTP in full plantar flexion with >100-110% BW added
Hip abduction, adduction, external rotation, and extension strength	Day 1 post surgery as symptoms allow	Open chain non weight bearing hip strengthening	% body weight as per discharge criteria
Foot/ankle strength (tibialis anterior/posterior, peroneals, flexor halluc longus)	Day 1 post surgery as symptoms allow	Open chain non weight bearing foot/ankle strengthening	Grade 5/5 muscle strength on manual testing

All streams progressed as symptoms and appropriate technical competency allow.



Single Leg Slant Board Calf Raise



Front Squat



Hip Abduction/External Rotation Banded

Clinical Pearls

Foot:

- Foot and ankle control significantly influences frontal plane knee and hip control. Targeting this early and progressively throughout rehabilitation optimises the restoration of lower limb biomechanics later in the rehabilitation process.
- Combine isolated strengthening exercises (such as those targeting the tibialis posterior and peroneal muscles) with motor control exercises (such as calf raises onto the first metatarsophalangeal joint and mini squats) to optimize midstance and toe-off mechanics.

Hip:

- Target both external rotation strength and hip abduction strength – exercise selection and execution is key.
- Add progressive overload to hip strengthening exercises – while weight is commonly added to quadriceps and hamstring strengthening exercises, it is often overlooked for hip exercises. For instance, gradually increase resistance bands or weights throughout the rehabilitation process.

Strength

Strength training naturally follows many of the motor control exercises outlined above. Once technical competency is achieved in a particular exercise, the intensity (weight) or volume (sets or repetitions) can be adjusted to achieve the desired training effect. It is crucial to ensure appropriate activation and recruitment of individual muscle groups, especially the quadriceps and hamstring muscle groups, before progressing to higher intensity strength training.

The primary precautions regarding strength training relate to the response of the graft donor site (commonly the patellar/quadriceps/hamstring) and the injured knee joint itself. Therefore, it is vital that all strength training exercises engage the targeted muscle groups without causing any sensation or irritation to the donor sites or the knee joint, both during the exercises and the following morning (highlighting the importance of daily monitoring, as outlined in the key performance indicators above). At Aspetar, strength training commences as early as possible in the rehabilitation process and is initiated by our physiotherapists. Transition to the strength team occurs once pain-free strengthening with proper technique of all major muscle groups has been achieved, facilitating progression to higher intensity exercises.

Secondary precautions regarding graft laxity, particularly concerning open-chain quadriceps exercises, are typically addressed by adhering to primary precautions of strengthening at intensities that do not aggravate the graft donor site or the knee joint. Additionally, research indicates that open-chain exercises do not negatively affect graft laxity when initiated between 4 to 6 weeks post-surgery³⁻⁵ which generally allows for loading at intensities that do not cause knee joint or graft donor site aggravation.

Regardless of the post-surgery stage, strength training programming should prioritize the individual athlete's competence and deficits, as identified through periodic (6-week) assessments.



Isokinetic Testing

Strength Training Progression

Muscle Group	Entry Point	Transitions				
Timeline	Day 1 post surgery as symptoms allow	Once technical competency achieved as symptoms allow	From 4-6 weeks, as symptoms allow	From 6-8 weeks, as symptoms allow	From 10 weeks, as symptoms allow	Once training > 5RM concentrically, as symptoms allow
Quadriceps	Isometrics (+/- EMG/NMES)	Closed chain (double/single leg squat /leg press)	Open chain modified (30°- 90°)	Open chain (low load full range occlusion can begin earlier as tolerated)	Isokinetic	Eccentric open chain
Hamstrings	Isometrics (+/- EMG/NMES)	Closed chain (deadlift/ single leg RDL)	Open chain modified (30°- 90°)	Open chain full range	Isokinetic	Eccentric open chain
Calf	Seated (knee flexed) and standing (knee extended)					

All streams progressed as symptoms and appropriate technical competency allow.

Strength training should be broadly divided into three phases:

- **General Strength Training** (12-15 repetitions, 3-4 sets) – Initial progression of intensity and volume once technical competency in an exercise is achieved.
- **Hypertrophy** (8-12 repetitions, 4-6 sets) – Targeted development of muscle growth in individual limbs or muscle groups, complemented by occlusion training if needed.
- **Max Strength** (< 5 repetitions, 4-6 sets) – Targeting changes in the central and peripheral nervous system to maximise force production.

All these strength training phases are necessary at different stages of the rehabilitation process to ensure appropriate progression without causing irritation to the donor site or knee joint while restoring muscle mass and function. Exercise selection should address deficits in specific muscle groups and ranges of motion related to the initial injury and subsequent surgery, resolving them to desired levels to meet discharge criteria.

Clinical Pearls

Quadriceps:

- The greatest peak torque deficits are at deeper ranges of knee flexion (i.e., $>90^\circ$). Strengthen the muscles through greater angles of knee flexion as tolerated.
- Peak torque measurements may fail to capture activation and strength deficits in the inner range of quadriceps. Always focus on targeting inner range activation and peak torque development simultaneously.

Hamstrings:

- Hamstring weakness, particularly in the medial hamstring, is always greater after ACLR using hamstring graft. Target accordingly with inner range knee flexion exercises to bias the medial hamstring.
- Combine hip and knee dominant hamstring exercises throughout rehabilitation to target the entire muscle group and function.



Rectus Femoris Strength



Closed Chain Strength



Isolated Quadriceps Strength

Explosiveness

Explosiveness relates to the ability to produce force concentrically or eccentrically in relation to time (i.e. how much force can be produced in the shortest duration possible). It is a key component of athletic development and performance, and a common deficit after ACL injury and reconstruction. Progression to jump and landing exercises should be based on the ability to execute the selected exercises without pain and with proper technique, considering early weight-bearing restrictions.

The primary precaution revolves around the response of the graft donor site and the knee joint, particularly in cases of concurrent chondral or meniscal injury. Graft laxity becomes a secondary concern once knee joint response is closely monitored. Research indicates that drop landing from a 60cm height induces lower ACL strain, ACL force, and anterior tibial shear compared to level ground walking.⁶

It is also worth noting that jumping is not the only method of developing explosiveness. All strength training can improve explosiveness (reported as rate of force development) as long as there is a clear intention to contract or move as fast as possible during exercise execution.⁷ Hence, concentric rate of force development can be initiated prior to jumping providing the intent to contract as fast as possible during strength training is consistently emphasized.

Explosive Strength Training Progression

Exercise Streams	Entry Point Criteria	Suggested Entry Exercise	Minimal Exit Point
Double leg landing	Front squat to 90° knee flexion	Double leg drop catch from double leg tip toe position	Maximal double leg CMJ
Single leg landing	Single leg squat to 90° knee flexion	Single leg drop catch from double leg tip toe position	Maximal single leg jump in multiple planes
Double leg jumping	Having commenced single leg landing	Band supported double leg jumping	Maximal double leg CMJ
Single leg jumping	Having commenced submaximal multiplanar single leg landing	Band supported Single leg jumping	Maximal single leg jump in multiple planes

All streams progressed as symptoms and appropriate technical competency allow.

Clinical Pearls

Concentric:

- Rate of force development can be improved during strength training without involving jumping or leaving the floor, particularly in the early stages of rehabilitation. The key is to ensure that the intended concentric contraction is executed as fast as possible or as tolerated.
- During jump training, emphasis should be placed on both technique and intensity. External cues such as "touch the ceiling" can facilitate the redevelopment of triple extension co-ordination throughout the kinetic chain.

Eccentric:

- When aiming to improve eccentric capacity, focus on both technique and intensity. This includes coaching and challenging deeper ranges of knee flexion, which are common deficits in jump and change of direction testing.
- Include exercises with higher intensity, such as drop landings or weighted landings, as well as those that challenge motor control, such as directional hops with or without perturbations, to optimize outcomes.



Concentric Explosiveness



Eccentric Explosiveness

Reactive Strength

Deficits in reactive strength or plyometric ability are among the most persistent physical deficits post-ACLR.^{8,9} Plyometric ability and repetitive hopping are also key precursors to running. Plyometric exercises have been highlighted as a crucial component in effective ACL prevention programs.

While high-intensity plyometric exercises are demanding on the neuromuscular system and the knee joint, lower-intensity extensive plyometric exercises can, and should, be initiated early in rehabilitation to develop sufficient competency before returning to running (refer to the pogo competency requirement in the running section).

Reactive Strength Training Progression

Exercise Streams	Criteria for Introduction	Entry Point	Transition Point
Toe Taps	Once gait normalised	Alternate leg toe taps	1 min alternate leg toe taps
Pogos	>6 weeks	Banded alternate leg pogos	>20 Single leg pogos
Cone Hopping	> 12 weeks Single leg pogos complete	Line hopping	Forward/backward cone hopping
Tuck Jumps	Cone hopping complete	Double leg tuck jumps	> 5 single leg tuck jumps with knee to hip height
Drop Jumps	Tuck jumps complete	Double leg drop jump 30cm	Single leg drop jump 15cm

All streams progressed as symptoms and appropriate technical competency allow.

Technical competency to include no heel strike and minimal ground contact time

Clinical Pearls

Reactive Strength:

- Focus on developing short ground contact time early in rehabilitation. Many athletes may regain jump height later in rehabilitation but continue to experience asymmetries in ground contact time, which negatively impacts their reactive strength index.
-
- Focus on developing technique and intensity concurrently. Ensure that the heel is off the ground to target ankle stiffness. Gradually increase weight, distance, or height in plyometric exercises to progress intensity in accordance with technique and knee tolerance.



Single Leg Pogos



Single Leg Cone Hopping Forward/Back

Running

A key consideration when determining the criteria for return to running is to precisely define what constitutes a "run." As illustrated by the Aspetar ACL Protocol figure on page 16, the development of running mechanics begins very early post-surgery, well before the athlete embarks on their first "run." For our protocol, we have set the benchmark for return to running as 10km/h x 200m x 6 repetitions. This does not include any pool running or alter-G running that is carried out prior to this stage.

The criteria for return to running (as defined above):

- >12 weeks post-surgery
- No/trace effusion
- Full knee extension and 135° knee flexion
- 0/10 pain during running drills and no change in pain or swelling next morning
- Hydrotherapy running program complete (10km/h x 15 min at 30% BW)
- Alter-G running program complete (after hydrotherapy protocol - transition from 50 to 90% BW 10km/h x 15min)
- Single leg pogo with good form (heel off ground, knee extended) x 30 repetitions
- Neutral pelvis in frontal plane at midstance
- >70% quadriceps LSI
- IKDC > 64%¹⁰

The aim of the running program is to prepare the athlete to withstand the transition to sports specific training by gradually increasing running volume and intensity. This will naturally vary depending on the sport, but the outline provided below is tailored for field sport athletes (it may be adjusted to 12 km/h for court sports athletes). This guide serves as an example of how hydrotherapy/Alter-G/treadmill/field running could be transitioned. The key indicators of success are the daily KPI previously outlined in terms of pain, swelling, and ROM. Acceleration mechanics can be developed concurrently with the treadmill or field running program, and exposure to high speeds can be included once the knee joint and graft donor site can tolerate the volume and intensity.



Running Assessment

Running Progression

	Hydrotherapy	Anti-Gravity Treadmill	Treadmill/Field Running
Entry Criteria	Skipping in pool	Completion of hydrotherapy protocol	Completion of the Alter-G protocol
Time Post-Surgery	From 7-8 weeks	From 10-12 weeks	From 12 weeks
Entry Session	6-7km/h x 30% BW x 10 min	10km/h x 50% BW x 15 min	10km/h x 200m x 6 repetitions
Progression	Increase speed or time per session	Increase 10% BW per session	Increase 2km/h or 2 repetitions per session
Exit Session	10km/h x 30% BW x 15 min	10km/h x 90% BW x 15 min	16km/h x 200m x 8 repetitions 12km/h for court sports and some recreational athletes

All streams progressed as symptoms and appropriate technical competency allow.



Banded Acceleration

Clinical Pearls

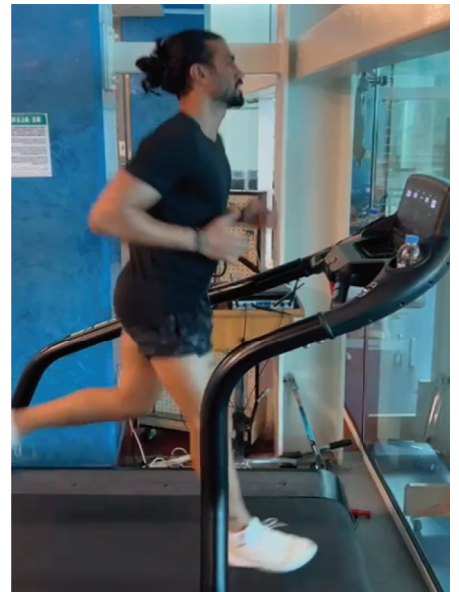
- Prioritise the development of running mechanics before initiating running for volume, and consistently advance the running mechanics drills throughout the rehabilitation process.
- Build up knee tolerance to cyclical loading in advance of increasing running volume through exposure to underwater treadmill, anti-gravity treadmill or band assisted pogos to ensure minimal reaction in the knee when running volume increases.



Hydrotherapy



Anti-Gravity Treadmill



Treadmill

Change of Direction

When to commence change of direction drills and what constitutes a change of direction drill can vary among clinicians depending on the angle and approach speed of the task. Change of direction mechanics can initiate well before athletes engage in directional changes with significant intensity. The eccentric component of the explosiveness stream (landing) serves as the starting point for change of direction, representing the fundamental competency to decelerate one's center of mass with appropriate mechanics before subsequent push-off. Similarly, the reintroduction of simple footwork coordination drills (such as lateral push-off, lateral crossover, backwards and side shuffles) can, and should, commence well in advance of higher-intensity change of direction drills.

For the purposes of this protocol, **change of direction is defined as running into a side step with a change of direction angle greater than 45°**. In our practice, this would commence on the field as part of sports specific training, with simpler, lower intensity mechanical drills done in advance of this in the gym. This is distinguished from agility, which involves changing direction in response to an external stimulus (such as external cueing/instruction or opponent movement). Agility training occurs in the later stages of sports specific training once sufficient development of change of direction mechanics has been achieved.

The criteria for initiating sports specific/change of direction drills (as defined above) are as follows:

- From 18 weeks post-surgery
- Single Leg Squat 90° with neutral alignment
- Forward/backward cone hopping
- Neutral landing mechanics in submaximal hop in all three planes (forward/backward, medial/lateral, 90° clockwise/anti-clockwise)
- Running program completed (16km/h x 200m x 8 repetitions) without reaction.
- Symmetrical lateral push off and crossover step
- > 80% LSI in isokinetic knee strength
- >80% LSI in concentric and eccentric impulse during double CMJ and DJ
- >80% LSI in RSI during SLDJ

The athlete should be able to carry out maximal intensity planned efforts at multiple angles (i.e. 45°, 90°, 180°) with appropriate and symmetrical mechanics prior to transitioning to agility and higher intensity open sports specific drills.



Change of Direction Assessment

Clinical Pearls

- Add constraints into the exercises (e.g., holding a stick overhead to reduce trunk sway or using lateral band pulls to enhance push-off) to optimize the desired changes within and between sessions.
- Ensure that reactive strength and eccentric explosive components are thoroughly developed and progressed throughout this phase to prevent them from contributing to biomechanical deficits during change of direction drills.



Change of Direction Training

Sports Specific

There is naturally significant variation in the sports specific component of ACL rehabilitation, depending on the sport and the athlete's level of participation. While the entry point for sports specific training, particularly for field sports, may be relatively consistent, the duration and progression will largely depend on the sport's demands and the required training volume and intensity level necessary to return to training and competition at pre-injury levels. Consequently, we provide a brief overview of the considerations for field sports.

At each stage of the process, there is a gradual increase in intensity and demand, while simultaneously integrating sports specific skills. Running volume and intensity are closely monitored for each session, along with the knee's response the following day (following the daily Key Performance Indicators previously outlined). Consistent with the fundamental principles of the rehabilitation process, all sessions are individualized and progressed according to the athlete's specific needs and responses.

Stage 1 focuses primarily on linear running drills incorporating ball where appropriate.

Stage 2 advances to multidirectional movements incorporating ball where appropriate.

Stage 3 progresses the intensity and demand of both linear and multidirectional drills.

Stage 4 & 5 then progresses to incorporate all drills specific to their sport and finishes with contested training simulations.



Virtual Reality Training

Example of a Sports Specific Program for Football

	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Goal	Linear movements	Multidirectional movements	High-intensity linear and multidirectional combined	Football specific movement & skill restoration	Training simulation
Components	Linear running	Multidirectional running	High-intensity linear and multidirectional combined	Football specific skills (crossing shooting)	Football specific skills (crossing dribbling shooting)
	Acceleration-deceleration	Acceleration-deceleration	Passing & dribbling	Passing & dribbling	Passing & dribbling
	Passing & dribbling	Passing & dribbling	Football specific skills (corner crossing)	Contested drills	High-intensity sided
	Reactive drills	Reactive drills	Reactive drills	High-speed running (shooting under fatigue)	Real game scenarios
	Running load & conditioning	Running load & conditioning	Running load & conditioning		High-speed running (shooting under fatigue)
					Position Specific
Frequency	x 2 per week	x 2 per week	x 2 per week	x 3 per week	x 3 per week

All streams progressed as symptoms and appropriate technical competency allow.

These sessions can be supplemented with aerobic or interval running at the conclusion of the sports specific session if additional conditioning or running volume is necessary to advance towards suitable levels for transitioning back to their sport and level of participation.

Clinical Pearls

- Combine gradual exposure to sports specific skills and demands while persistently challenging the mechanics developed in running and change of direction components.
- Once the fundamental skills and movements for a sport have been reintroduced, further challenge these skills to align with the specific demands of the position played (e.g., defender vs midfielder in football; circle runner vs winger in handball).



Sports Specific Training

■ Conditioning

The early redevelopment of cardiovascular fitness and sports specific conditioning is crucial for both the physical and psychological recovery of athletes throughout the rehabilitation process and their successful return to optimal sporting performance after discharge.

Assessment of conditioning and the strategies employed to enhance it depend on the stage of rehabilitation and the knee's response to progressions. As rehabilitation nears completion, there may be overlap between sports specific and reconditioning sessions, which can be combined or complementary depending on the sport and the extent of reconditioning needed for each athlete's specific requirements in their sport.

Conditioning Progression and Testing

Time Post Surgery	Criteria	Potential Modalities	Assessment
3-4 weeks	Upper body	Upper body ergometer Ski ergometer sitting Rowing machine no legs	Upper body cycle test
>7 weeks	Non impact	As above Ski ergometer standing Wattbike Elliptical/cross trainer Versaclimber* Jacobs ladder*	Upper body cycle test
>12 weeks	Low impact	As above Stepper Stairmaster Air runner	Bike test
>18 weeks	On feet	As above Running	Yo-Yo test
> 24 weeks	On field	As above Sports specific drills	Yo-Yo test

* in latter weeks of this phase, once other exercises are well tolerated

All streams progressed as symptoms and appropriate technical competency allow.

Clinical Pearls

- Introduce conditioning as early as possible in the rehabilitation process, as it offers numerous mental and physical benefits crucial to the rehabilitation journey.
- As the athlete's running volume increases, transition the conditioning program to prepare them for the specific demands of their sport, such as total running distance, high-speed meters, maximum speed distance/exposures etc.



Watt Bike



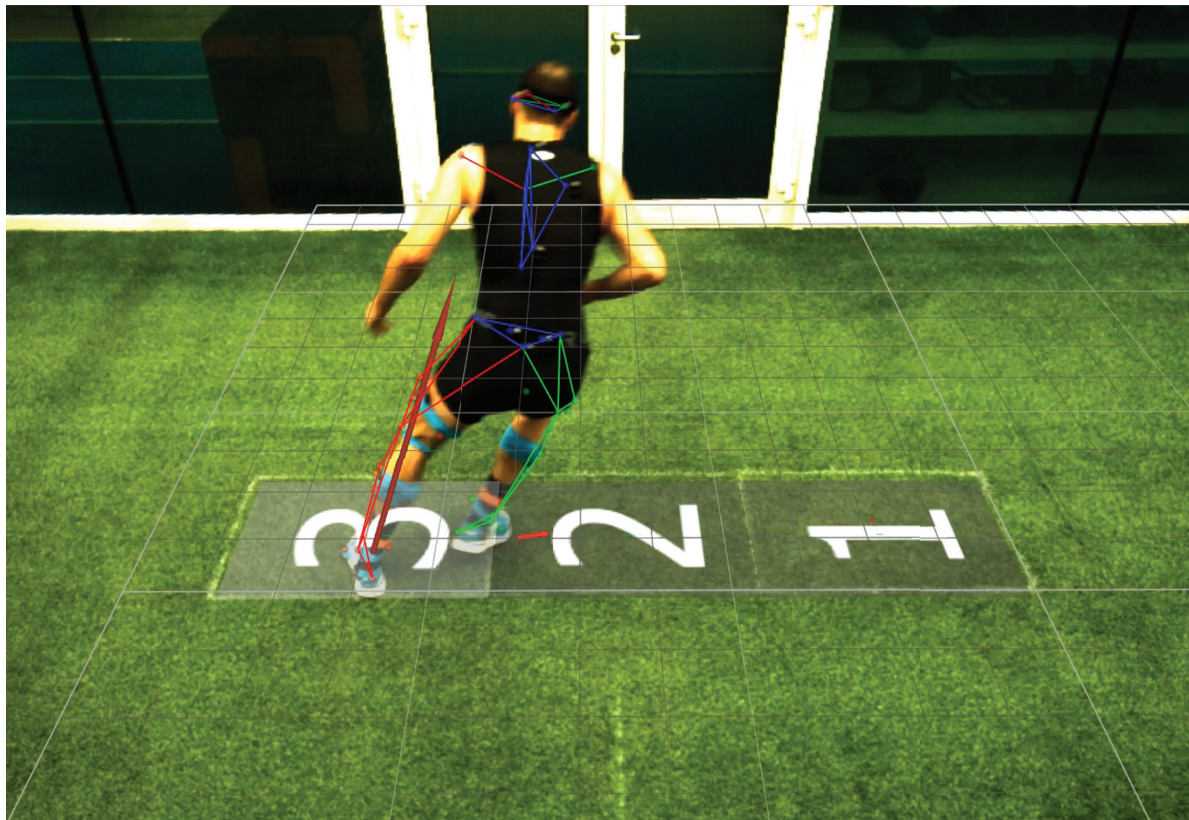
Jacobs Ladder

Discharge Criteria

There can be considerable variation in the criteria used to determine when an athlete should be discharged after ACL injury. The primary goal of the rehabilitation process is to restore the athlete to their desired level of sporting performance, ideally surpassing it, by fully restoring physical function to baseline levels or higher, and addressing any modifiable risk factors influencing subsequent injury. These criteria must be met before the athlete can return to unrestricted involvement in their pre-injury sport, which includes full participation in training sessions or equivalent activities.

The level of physical competency required will depend on the athlete's baseline function and their desired activity level post-rehabilitation. Consequently, discharge criteria should be tailored to the individual athlete's goals and aspirations. While there are subcategories, the criteria can generally be grouped into two main categories:

- Competitive Athlete Populations – Playing pivoting sports competitively
- Recreational Athlete Populations – Playing pivoting sports recreationally



3D Biomechanical Analysis

Competitive Athletes

Category	Test	Variables	Absolute Threshold	Symmetry Threshold
Motor Control	Single leg squat	Knee flexion	>90° knee flexion with neutral frontal plane control throughout the kinetic chain	>90%
Strength	Hip abduction	Peak eccentric force	>40% BW	>90%
	Hip adduction	Peak eccentric force	>40% BW	>90%
	Hip external rotation	Peak isometric force	>30% BW	>90%
	Ankle plantar-flexors	Peak isometric force	>200% BW	>90%
	Quadriceps	Peak isokinetic concentric torque	>300% BW	>90%
	Hamstring	Peak isokinetic concentric torque	>175% BW	>90%
Explosiveness	Double and single leg countermovement jump	Concentric impulse	Baseline or sports average	>90%
		Eccentric impulse	Baseline or sports average	>90%
		Jump height	Baseline or sports average	>90%
		Peak landing force	Baseline or sports average	>90%
		Knee angles and moments	Baseline or sports average	>90%
		Ground reaction forces and joint power	Baseline or sports average	>90%
	Single leg hop for distance	Hop distance	Baseline or sports average	>90%
		Knee angles and moments	Baseline or sports average	>90%
		Ground reaction forces and joint power	Baseline or sports average	>90%
Reactive Strength	Double and single leg drop jump	Concentric impulse	Baseline or sports average	>90%
		Eccentric impulse	Baseline or sports average	>90%
		Peak landing force	Baseline or sports average	>90%
		Contact time	<0.3s DL; <0.35s SL	>90%
		Jump height	Baseline or sports average	>90%
		Reactive strength index	<1.5 DL; >0.5 SL	>90%
		Knee angles and moments	Baseline or sports average	>90%
		Ground reaction forces and joint power	Baseline or sports average	>90%
Running	16km/h treadmill	Knee angles and moments	Baseline or sports average	>90%
		Ground reaction forces and joint power	Baseline or sports average	>90%
		Hip/pelvic frontal plane biomechanics midstance	Neutral	Symmetrical
Change of Direction	Planned and unplanned 90°	Knee angles and moments	Baseline or sports average	>90%
		Ground reaction forces and joint power	Baseline or sports average	>90%
Conditioning	Yo-Yo (or equivalent sports specific conditioning test)	Distance covered	Baseline or sports average	n/a
		Heart rate recovery	Baseline or sports average	n/a
Sports Specific Training	Completion of sports specific training program	Cleared all game based scenarios in closed and open drills	No knee reaction and baseline technical competency	n/a

Recreational Athletes

Category	Test	Variables	Absolute Threshold	Symmetry Threshold
Motor Control	Single leg squat	Knee flexion	>90° knee flexion with neutral frontal plane control throughout the kinetic chain	>90%
Strength	Hip abduction	Peak eccentric force	>30% BW	>90%
	Hip adduction	Peak eccentric force	>30% BW	>90%
	Hip external rotation	Peak isometric force	>25% BW	>90%
	Ankle plantar-flexors	Peak isometric force	>150% BW	>90%
	Quadriceps	Peak isokinetic concentric torque	>260% BW	>90%
	Hamstring	Peak isokinetic concentric torque	>160% BW	>90%
Explosiveness	Double and single leg countermovement jump	Concentric impulse	Baseline or sports average	>90%
		Eccentric impulse	Baseline or sports average	>90%
		Jump height	Baseline or sports average	>90%
		Peak landing force	Baseline or sports average	>90%
Reactive Strength	Double and single leg drop jump	Concentric impulse	Baseline or sports average	>90%
		Eccentric impulse	Baseline or sports average	>90%
		Peak landing force	Baseline or sports average	>90%
		Contact time	<0.3s DL; <0.35s SL	>90%
		Jump height	Baseline or sports average	>90%
		Reactive strength index	<1.3 DL; >0.4 SL	>90%
Running	16km/h treadmill	Vertical ground reaction force	Baseline or sports average	>90%
		Knee flexion and valgus angle	Baseline or sports average	>90%
		Hip/pelvic frontal plane biomechanics midstance	Neutral	Symmetrical



Clinical Pearls

- Educate the athlete on the testing process and the progression of the tests throughout the rehabilitation process either before or just after surgery. This ensures they understand what lies ahead and how each component of the program is assessed and progressed based on the results.
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- Periodise when testing is carried out to ensure the athlete has had sufficient time to adapt to the training stimulus. This ensures that their testing results best represent their current physical status, their progress, and appropriately highlight the areas to be prioritized in the next phase of rehabilitation.

References

1. Kotsifaki R, Korakakis V, King E, et al. Aspetar clinical practice guideline on rehabilitation after anterior cruciate ligament reconstruction. *British journal of sports medicine* 2023;57(9):500-14.
2. King E. Rehabilitation after ACL reconstruction; the Aspetar way. *Aspetar sports medicine journal*, 2023;12(TT29):284-290.
3. Forelli F, Barbar W, Kersante G, et al. Evaluation of Muscle Strength and Graft Laxity With Early Open Kinetic Chain Exercise After ACL Reconstruction: A Cohort Study. *Orthopaedic journal of sports medicine* 2023;11(6):23259671231177594.
4. Perriman A, Leahy E, Semciw AI. The Effect of Open- Versus Closed-Kinetic-Chain Exercises on Anterior or Tibial Laxity, Strength, and Function Following Anterior Cruciate Ligament Reconstruction: A Systematic Review and Meta-analysis. *The Journal of orthopaedic and sports physical therapy* 2018;48(7):552-66.
5. Wright RW, Preston E, Fleming BC, et al. A systematic review of anterior cruciate ligament reconstruction rehabilitation: part II: open versus closed kinetic chain exercises, neuromuscular electrical stimulation, accelerated rehabilitation, and miscellaneous topics. *Journal of knee surgery* 2008;21(3):225-34.
6. Escamilla RF, Macleod TD, Wilk KE, et al. Anterior cruciate ligament strain and tensile forces for weight-bearing and non-weight-bearing exercises: a guide to exercise selection. *The Journal of orthopaedic and sports physical therapy* 2012;42(3):208-20.
7. Blazeovich AJ, Wilson CJ, Alcaraz PE, Rubio-Arias JA. Effects of resistance training movement pattern and velocity on isometric muscular rate of force development: a systematic review with meta-analysis and meta-regression. *Sports medicine* 2020;50(5):943-963.
8. King E, Richter C, Franklyn-Miller A, et al. Back to normal symmetry? Biomechanical variables remain more asymmetrical than normal during jump and change-of-direction testing 9 months after anterior cruciate ligament reconstruction. *The American journal of sports medicine* 2019;47(5):1175-85.
9. Kotsifaki A, Van Rossom S, Whiteley R, et al. Single leg vertical jump performance identifies knee function deficits at return to sport after ACL reconstruction in male athletes. *British journal of sports medicine* 2022;56(9):490-98.
10. Pairot de Fontenay B, Van Cant J, Gokeler A, et al. Reintroduction of Running After Anterior Cruciate Ligament Reconstruction With a Hamstrings Graft: Can We Predict Short-Term Success? *Journal of athletic training* 2022;57(6):540-46.

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